

Web Accessibility



What is *Accessibility*?

The W3C promotes the *Web Accessibility Initiative* (WAI)

<https://www.w3.org/WAI/>



According to the W3C, *Web accessibility* means that “people with disabilities can use the Web”

Everybody should be able to access the Web

Usability vs. Accessibility. Strictly correlated, but:

- Usability has more to do with “logic” and “understanding”
Normally, usability cannot be evaluated automatically
- Accessibility (although **not always**) has more to do with technical aspects and well-defined rules

WCAG 1.0

14 guidelines (WCAG = Web Content Accessibility Guidelines)

structured into “checkpoints”, defined in 1999

(<https://www.w3.org/TR/WCAG10/>)

They have been the basis for several web accessibility standards worldwide

WCAG 2.0

Guidelines based on WCAG 1.0, but defined according to a different “philosophy”

They are not “technique-centered” but “principle-centered”
(2008, <https://www.w3.org/TR/WCAG20/>)

4 *principles*:

- Perceivable
- Operable
- Understandable
- Robust

Each principle has *success criteria*

WCAG 2.0 – Principles

Principle 1: *Perceivable*

Information and user interface components must be presentable to users in ways they can perceive

Principle 2: *Operable*

User interface components and navigation must be operable

Principle 3: *Understandable*

Information and the operation of the user interface must be understandable

Principle 4: *Robust*

Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies

WCAG 2.0 – Guideline 1.1

Text Alternatives: Provide text alternatives for any non-text content so that it can be changed into other forms people need, such as large print, braille, speech, symbols, or simpler language

Success Criterion 1.1.1 Non-text Content: All non-text content that is presented to the user has a text alternative that serves the equivalent purpose, except for the situations listed below (Level A)

- Controls, Input: If non-text content is a control or accepts user input, then it has a name that describes its purpose
- ...
- Decoration, Formatting, Invisible: If non-text content is pure decoration, is used only for visual formatting, or is not presented to users, then it is implemented in a way that it can be ignored by assistive technology

WCAG 2.1 and 2.2

Extend WCAG 2

E.g. (2.1), Guideline *1.3 Adaptable*, *1.3.4 Orientation* (AA):

Content does not restrict its view and operation to a single display orientation, such as portrait or landscape, unless a specific display orientation is essential

Web Content Accessibility Guidelines (WCAG) 2.1

W3C Recommendation 05 June 2018



Web Content Accessibility Guidelines (WCAG) 2.2

W3C Recommendation 05 October 2023



USA – Rehabilitation Act and Section 508

Rehabilitation Act (1973)

Federal law that prohibits discrimination on the basis of disability in programs conducted by Federal agencies, in programs receiving Federal financial assistance, in Federal employment, and in the employment practices of Federal contractors



Section 508 (1998)

An amendment to the Rehabilitation Act that establishes requirements for electronic and information technology developed, maintained, procured, or used by the Federal government. Section 508 requires Federal electronic and information technology to be accessible to people with disabilities, including employees and members of the public

Italy – *Legge Stanca* (*Stanca Act*)

January 9th, 2004

Act issued (“Legge 04/2004”, also known as “Legge Stanca”)



March 1st, 2005

Application decree by the President of the Republic

July 8th, 2005

Ministerial decree on the “Technical requirements and the different levels of accessibility of computer tools”

August 8th, 2005

Ministerial decree published on the “Gazzetta Ufficiale” (Italian official journal) N. 183

<https://www.agid.gov.it/it/design-servizi/accessibilita/normativa>

Assistive technologies

Depend on the kind of disability

- Vision
 - Motion
 - Hearing
 - Cognitive
- } main

Require proper page design

Each kind of *user agent* has its own features

Are also useful/necessary for elderly people

Vision declines, movement difficulties arise... we all hope to grow old!



Vision disabilities

Low vision

Screen magnifiers

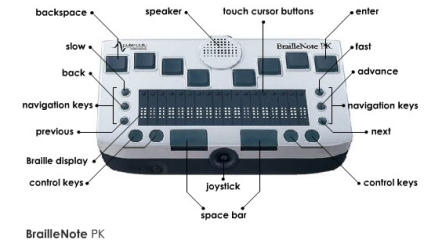


Blindness

- Screen readers, e.g.:
 - NVDA (open source, <https://www.nvaccess.org/>)
 - JAWS (<https://vispero.com/jaws-screen-reader-software/>)



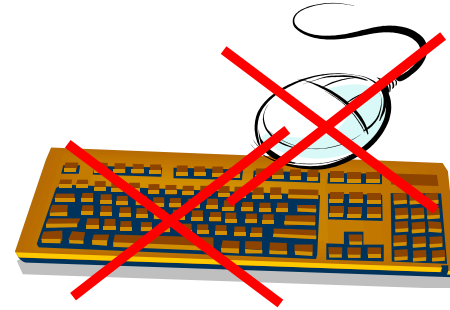
- Braille readers
 - Touch devices (especially useful for deaf-blind people)



Motion disabilities

Depending on the level of disability:

- Mouse pointing difficulties
- Inability to use the mouse
- Inability to use the keyboard



Solutions:

- Make the website independent of (or not totally dependent on) the mouse
- Speech recognition
- Use of special keyboards
- Use of mouth sticks or head wands
- Use of eye trackers



Text

Specify the language

E.g., `<html lang="it">`, `<span lang="en">file</span>`

Use meaningful titles for pages

`<title>` tag; people using a screen reader will be able to know in advance whether a page provides useful information...

Use tags correctly (from a semantic point of view)

Use clear language

It eases content understanding

Font size

Use relative sizes

The user should be able to enlarge fonts

Segue un esempio di come appare il contenuto del file. In questo caso, l'elemento <GREETING> è stato personalizzato per visualizzare il contenuto in rosso, centrato all'interno del browser e con un carattere di 36 punti. L'elemento <MESSAGE> è stato personalizzato per visualizzare il testo in nero con un carattere di 18 punti. La parte display: block indica che il contenuto di questi elementi dovrà essere visualizzato in un blocco, in questo caso visualizzato su singole righe. È possibile vedere il risultato in due browser che supportano XML, nelle Figure 1.3 e 1.4. La Figura 1.3 mostra il documento visualizzato con il browser (disponibile solo in una versione preliminare all'epoca della pubblicazione di questo libro) e la Figura 1.4 mostra lo stesso documento visualizzato con Internet Explorer.

Big improvement for low-vision people!

Segue un esempio di come appare il contenuto del file. In questo caso, l'elemento <GREETING> è stato personalizzato per visualizzare il contenuto in rosso, centrato all'interno del browser e con un carattere di 36 punti. L'elemento <MESSAGE> è stato personalizzato per visualizzare il testo in nero con un carattere di 18 punti. La parte display: block indica che il contenuto di questi elementi dovrà essere visualizzato in un blocco, in questo caso visualizzato su singole righe. È possibile vedere il risultato in due browser che supportano XML, nelle Figure 1.3 e 1.4. La Figura 1.3 mostra il documento visualizzato con il browser (disponibile solo in una versione preliminare all'epoca della pubblicazione di questo libro) e la Figura 1.4 mostra lo stesso documento visualizzato con Internet Explorer.

Color contrast

Use contrasting colors for text and background

With sufficiently high differences in *luminance* and *chrominance*

The W3C has defined specific algorithms

- $\text{Luminance} = ((R * 299) + (G * 587) + (B * 114)) / 1000$

Rule: $| \text{background lumin.} - \text{foreground lumin.} | > 125$

- $\text{Chrominance} = [\text{Max}(R1, R2) - \text{Min}(R1, R2)] +$
 $[\text{Max}(G1, G2) - \text{Min}(G1, G2)] +$
 $[\text{Max}(B1, B2) - \text{Min}(B1, B2)]$

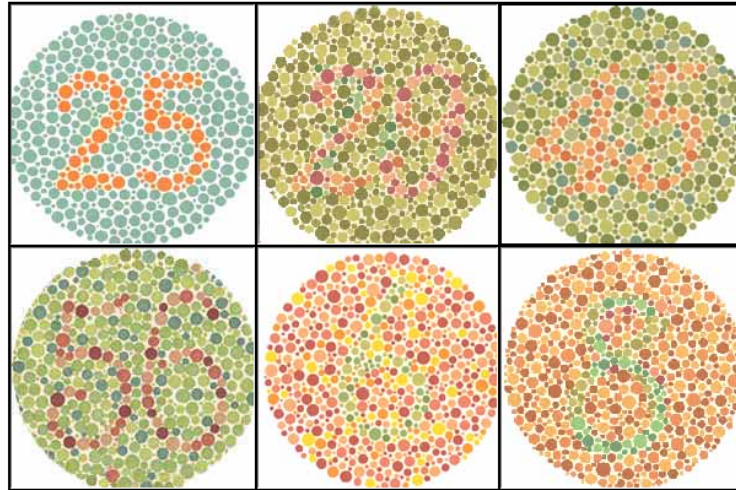
Rule: $\text{chrominance} > 500$

Color-based information

Do not rely only on color to convey information

Color blindness problems are very common (~ 1 in 12 males and 1 in 200 females in the world)

Test:



Images and multimedia content in general

If the multimedia element is important to understand the content, provide a textual description

For images, use the `alt` attribute (if the purpose is only “decorative”, `alt=""`)

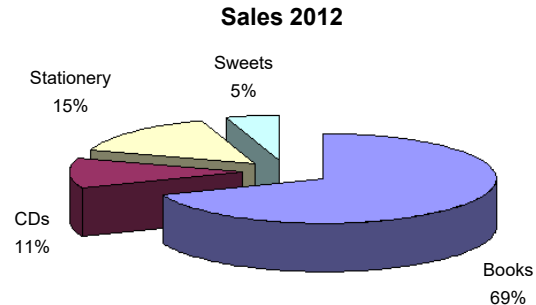
The alternative text **is not a caption!**

A caption is for those who can see...

E.g., “*Chart of sales in 2012*”

or

“*Sales in 2012: books 69%, CDs 11%, stationery 15%, sweets 5%*” ?



Navigation

Use meaningful words for links

Avoid the “click here” link: it does not provide any information when read (for example) by a screen reader!

Add titles to links

`title` attribute

E.g., `<a href="b.html" title="Biography of...">...</a>`

Be careful when using JavaScript links

A screen reader may not properly support them

Avoid opening new windows

They may be misleading (especially when using screen readers)

Navigation shortcuts

Provide (hidden) links to the important parts of the page

E.g., CSS: `.hidden {position: relative; left: -10000px}`
HTML: `<a class="hidden" href="#content">`
 Go to content
 `</a>`

Make links and form fields directly selectable through the keyboard

`accesskey` attribute

E.g., `<a href="p2.html" accesskey="2">Page 2</a>`
 `<input type="text" id="date" accesskey="d">`

Audio and video

For deaf people:

Provide transcripts for speech audio and captions for videos

For blind people:

- Provide text or audio descriptions for videos
- Depending on the kind of video, it may be necessary or useful to also describe the “scene” in which an action takes place

Some rules for accessible design

Moreover...

Other accessibility rules are related to:

- Tables
- Forms
- Client-side and server-side technologies
- Plugins
- ...

Some accessibility evaluation tools

Some tools

Wave

wave.webaim.org

Accessibility Checker

www.accessibilitychecker.org

Web Accessibility Evaluation Tools List (by W3C)

www.w3.org/WAI/ER/tools

Web accessibility is important!

*“The power of the Web is in its universality.
Access by everyone regardless of disability
is an essential aspect”*

Tim Berners-Lee
(W3C Director and inventor of the World Wide Web)